LABSHEET-5

Task 1: Implement Queue Using Write a C++ program to implement a Queue using an array and perform the following operations:

□ Enqueue an element into the queue.

□ Dequeue an element from the queue.

□ Display the front element.

□ Display all elements of the queue.

□ Check whether the queue is empty or full.

```
#include <iostream>
using namespace std;
class Queue {
private:
   int arr[SIZE];
   int front;
   int rear;
public:
   Queue() {
      front = -1; rear = -1;
   }
   bool isEmpty() {
      return (front == -1);
   }
   bool isFull() {
      return (rear == SIZE - 1);
   }
   void enqueue(int value) {
      if (isFull()) {
         cout << "Queue Overflow! Cannot insert " << value << ".\n";
         return; }
      if (front == -1) {
```

NAME-ANSHIKA GULATI
BRANCH-CSE(AIML)
CU26220150
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```
        front = 0;
    }
    rear++;
    arr[rear] = value;
    cout << value << " inserted into queue.\n";
}
void dequeue() {
    if (isEmpty()) {
        cout << "Queue Underflow! Queue is empty.\n";
        return;
    }
    cout << arr[front] << " deleted from queue.\n";
    if (front == rear) {
        front = -1;
        rear = -1;
    } else {
        front++;
    }
}
void displayFront() {
    if (isEmpty()) {
        cout << "Queue is empty.\n";
    } else {
        cout << "Front element: " << arr[front] << endl;
    }
}
void display() {
    if (isEmpty()) {
        cout << "Queue is empty.\n";
        return;
```

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```
        }
        cout << "Queue elements: ";
        for (int i = front; i <= rear; i++) {
            cout << arr[i] << " ";
        }
        cout << endl;
    }
};
int main() {
    Queue q;
    int choice, value;
    do {
        cout << "\n=== QUEUE MENU ===\n";
        cout << "1. Enqueue (Insert)\n";
        cout << "2. Dequeue (Delete)\n";
        cout << "3. Display Front\n";
        cout << "4. Display All Elements\n";
        cout << "5. Is Queue Empty?\n";
        cout << "6. Is Queue Full?\n";
        cout << "7. Exit\n";
        cout << "Enter your choice: ";
        cin >> choice;
        switch (choice) {
            case 1:
                cout << "Enter value to insert: ";
                cin >> value;
                q.enqueue(value);
                break;
            case 2:
                q.dequeue();
```

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```
            break;
        case 3:
            q.displayFront();
            break;
        case 4:
            q.display();  break;
        case 5:
            if (q.isEmpty()) cout << "Yes, the queue is empty.\n";
            else cout << "No, the queue is not empty.\n";
            break;
        case 6:
            if (q.isFull()) cout << "Yes, the queue is full.\n";
            else cout << "No, the queue is not full.\n";
            break;
        case 7:
            cout << "Exiting program. Goodbye!\n";
            break;
        default:
            cout << "Invalid choice! Please choose between 1 and 7.\n";
    }
  } while (choice != 7);
  return 0;
}
```

Task 2: Implement Queue Using Linked List Write a C++ program to implement a Queue using a singly linked list and perform the following operations:

□ Enqueue a new element into the queue.

□ Dequeue an element from the queue.

□ Display the front element.

□ Display all elements of the queue.

□ Check whether the queue is empty.

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```
#include<iostream>
using namespace std;
struct Node{
int data;
Node* next; };
class Queue {
private:
Node* front;
Node* rear;
public:
Queue(){
front=nullptr;
rear=nullptr;
}
bool isEmpty(){
return front==nullptr;
}
void enqueue(int value){
Node* temp=new Node();
temp->data=value;
temp->next=nullptr;
if(isEmpty()){
front=rear=temp;
}else{
rear->next=temp;
rear=temp;
}
cout<<"Element inserted successfully.\n";
}
void dequeue(){
```

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```
if(isEmpty()){
cout<<"Queue Underflow! Queue is empty.\n";
return;
}
Node* temp=front;
front=front->next;
if(front==nullptr){
rear=nullptr;
}
cout<<"Dequeued element: "<<temp->data<<"\n";
delete temp;
}
void displayFront(){
if(isEmpty()){
cout<<"Queue is empty.\n";
}else{
cout<<"Front element: "<<front->data<<"\n";
} }
void displayQueue(){
if(isEmpty()){
cout<<"Queue is empty.\n";
return; }
Node* temp=front;
cout<<"Queue elements: ";
while(temp!=nullptr){
cout<<temp->data<<" ";
temp=temp->next;
}
cout<<"\n";
}
```

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```
Queue(){
while(!isEmpty()){
Node* temp=front;
front=front->next;
delete temp;
} } };
int main(){
Queue q;
int choice,value;
do{
cout<<"\n--- Queue Menu ---\n";
cout<<"1. Enqueue\n";
cout<<"2. Dequeue\n";
cout<<"3. Display Front\n";
cout<<"4. Display All Elements\n";
cout<<"5. Check if Empty\n";
cout<<"6. Exit\n";
cout<<"Enter your choice: ";
cin>>choice;
switch(choice){
case 1:
cout<<"Enter value to enqueue: ";
cin>>value;
q.enqueue(value);
break;
case 2:
q.dequeue();
break;
case 3:
q.displayFront();
```

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```
break;
case 4:
q.displayQueue();
break;
case 5:
if(q.isEmpty()){
cout<<"Queue is empty.\n";
}else{
cout<<"Queue is not empty.\n";
}break;
case 6:
cout<<"Exiting program.\n";
break;
default:
cout<<"Invalid choice! Please try again.\n";
}
}while(choice!=6);
return 0;
}
```

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CU26220150
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LABSHEET-4

Task 1: Implement Stack Using Array Write a C++ program to implement a Stack using an array and perform the following operations:

- **Push an element into the stack.**
- **Pop an element from the stack.**
- **Peek/Top element of the stack.**
- **Display all elements of the stack.**
- **Check whether the stack is empty or full.**

```
#include <iostream>
using namespace std;
#define MAX 5
class Stack {
   int top;
   int arr[MAX];
public:
   Stack() { top = -1; }
   bool push(int x) {
      if (top >= (MAX - 1)) {
         cout << "Stack Overflow\n";
         return false;
      } else {
         arr[++top] = x;
         cout << x << " pushed into stack\n";
         return true;
      }
   }
   int pop() {
      if (top < 0) {
         cout << "Stack Underflow\n";
         return 0;
```

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BRANCH-CSE(AIML)
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```
        } else {
            int x = arr[top--];
            return x;
        }
    }
    int peek() {
        if (top < 0) {
            cout << "Stack is Empty\n";
            return 0;
        } else {
            return arr[top];
        }
    }
    bool isEmpty() {
        return (top < 0);
    }
    bool isFull() {
        return (top >= MAX - 1);
    }
    void display() {
        if (top < 0) {
            cout << "Stack is Empty\n";
        } else {
            cout << "Stack elements: ";
            for (int i = top; i >= 0; i--)
                cout << arr[i] << " ";
            cout << "\n";
        }
    }
};
```

NAME-ANSHIKA GULATI
BRANCH-CSE(AIML)
CU26220150
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```
int main() {
    Stack s;
    int choice, val;
    do {
        cout << "\n--- Stack Menu ---\n";
        cout << "1. Push\n2. Pop\n3. Peek\n4. Display\n5. Check Empty\n6. Check Full\n7. Exit\n";
        cout << "Enter your choice: ";
        cin >> choice;
        switch (choice) {
            case 1:
                cout << "Enter value to push: ";
                cin >> val;
                s.push(val);
                break;
            case 2:
                val = s.pop();
                if (val != 0) cout << "Popped element: " << val << "\n";
                break;
            case 3:
                val = s.peek();
                if (val != 0) cout << "Top element: " << val << "\n";
                break;
            case 4:
                s.display();
                break;
            case 5:
                if (s.isEmpty()) cout << "Stack is empty\n";
                else cout << "Stack is not empty\n";
                break;
            case 6:
```

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```
            if (s.isFull()) cout << "Stack is full\n";
            else cout << "Stack is not full\n";
            break;
        case 7:
            cout << "Exiting...\n";
            break;
        default:
            cout << "Invalid choice!\n";
      }
   } while (choice != 7);
   return 0;
}
```

Task 2: Implement Stack Using Linked List Write a C++ program to implement a Stack using a singly linked list and perform the following operations:

□ Push a new element into the stack.

□ Pop an element from the stack.

□ Peek/Top element of the stack.

□ Display all elements of the stack.

□ Check whether the stack is empty.

```
#include <iostream>
struct Node {
   int data;
   Node* next;
};
class Stack {
private:
   Node* top;
public:
   Stack() {
      top = nullptr;
```

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```
}
void push(int val) {
    Node* newNode = new Node();
    newNode->data = val;
    newNode->next = top;
    top = newNode;
}
void pop() {
    if (isEmpty()) {
        std::cout << "Stack Underflow\n";
        return;
    }
    Node* temp = top;
    top = top->next;
    delete temp;
}
int peek() {
    if (isEmpty()) {
        std::cout << "Stack is empty\n";
        return -1;
    }
    return top->data;
}
bool isEmpty() {
    return top == nullptr;
}
void display() {
    if (isEmpty()) {
        std::cout << "Stack is empty\n";
        return;
```

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```
        }
        Node* temp = top;
        while (temp != nullptr) {
            std::cout << temp->data << " ";
            temp = temp->next;
        }
        std::cout << "\n";
    }
    ~Stack() {
        while (!isEmpty()) {
            pop();
        }
    }
};

int main() {
    Stack s;
    s.push(10);
    s.push(20);
    s.push(30);
    std::cout << "Elements in stack: ";
    s.display();
    std::cout << "Top element: " << s.peek() << "\n";
    s.pop();
    std::cout << "After popping, elements: ";
    s.display();
    std::cout << "Is stack empty? " << (s.isEmpty() ? "Yes" : "No") << "\n";
    return 0;
}
```

NAME-ANSHIKA GULATI
BRANCH-CSE(AIML)
CU26220150
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